Semantic-Guided Autonomous Mapping: A Survey of Vision-Language-Action Models and Their Integration with Visual SLAM for Predictive Next-Best-View Exploration on Edge Devices

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***Abstract*—Autonomous exploration of complex infrastructures demands the seamless integration of high-fidelity spatial perception with cognitive reasoning capabilities. This survey systematically reviews the convergence of three rapidly evolving research threads: (i) visual Simultaneous Localisation and Mapping (ViSLAM) augmented with 3D Gaussian Splatting (3DGS) for real-time, photorealistic scene reconstruction; (ii) Vision-Language-Action (VLA) foundation models that endow robots with grounded spatial understanding and natural-language instruction following; and (iii) predictive Next-Best-View (Pred-NBV) planners that minimise exploration entropy under energy-constrained budgets. We articulate how VLA-driven semantic intelligence can proactively guide metric reconstruction through closed-loop Pred-NBV strategies, reducing flight effort while maximising surface coverage. Moreover, we analyse the feasibility of deploying these architectures on edge-computing hardware (e.g., NVIDIA Jetson) via parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) and 8-bit quantisation. A comparative analysis of reconstruction quality, exploration efficiency, cognitive performance, and hardware utilisation metrics consolidates the state of the art and identifies open challenges toward fully autonomous, semantically aware mobile robots and unmanned aerial vehicles (UAVs).**

***Index Terms*—survey, visual SLAM, 3D Gaussian Splatting, vision-language-action models, next-best-view planning, edge computing, autonomous exploration, UAV**

I. Introduction

Autonomous inspection of built environments requires robotic platforms capable of constructing dense, geometrically faithful maps while reasoning about *what* to observe next and *why*. Traditional pipelines decouple spatial perception from high-level cognition: a SLAM front-end densifies the map, and a separate planner selects viewpoints purely on the basis of geometric information gain [1]. This decoupling introduces two critical limitations: the planner remains "blind" to semantic context, and the computational cost of both dense reconstruction and large vision-language models has historically confined these systems to cloud-based deployments, precluding real-time autonomy on UAVs.

Recent advances motivate a re-examination of this architecture. 3D Gaussian Splatting (3DGS) surpasses NeRF in efficiency for robotic applications [2], [3]. Vision-Language-Action (VLA) models—such as those built on Qwen-VL [4] and Lumo-1 [5]—interpret complex instructions with spatial awareness, achieving success rates above 90% in task switching [6]. Predictive NBV algorithms observe up to 25.46% more surface points within the same budget [1], [7].

This paper fills the gap of a unified survey by:

1) Providing a taxonomy of modern VLA models and their coupling with real-time 3DGS-based SLAM systems (e.g., MCGS-SLAM [8], MVS-GS [9]).
2) Analysing computational feasibility on embedded hardware through LoRA and quantisation [10].
3) Proposing a closed-loop architecture unifying dense perception (3DGS), semantic cognition (VLA), and active exploration (Pred-NBV / GenNBV).

II. Background and Related Work

A. Visual SLAM and 3D Reconstruction

Visual SLAM constitutes the perceptual backbone of autonomous mobile robots. Multi-camera frameworks such as MCGS-SLAM [8] fuse observations from multiple viewpoints into a unified Gaussian map, while MVS-GS [9] combines online multi-view stereo with Gaussian-based scene representations. Key evaluation metrics include Absolute Trajectory Error (ATE) for localisation, Chamfer Distance (CD) for geometric fidelity, and rendering indicators such as PSNR, SSIM, and LPIPS [3].

B. 3D Gaussian Splatting

3DGS represents scenes as collections of 3D Gaussians, enabling real-time novel-view synthesis at rates exceeding 100 fps—and up to 200 fps in optimised configurations [11]. In urban-scale scenes, 3DGS achieves PSNR values of up to 28.93 dB, SSIM scores of 0.918, and LPIPS as low as 0.056, outperforming NeRF baselines [3], [12]. Liu et al. [2] trace the evolution from classical multi-view geometry through NeRF to 3DGS.

C. Vision-Language-Action Models

VLA models extend large vision-language models by incorporating an action output head, mapping visual observations and language instructions to robot control commands. Qwen2.5-VL [4] achieves spatial reasoning accuracies between 75.6% and 86.3% on the EmbSpatial benchmark. Lumo-1 [5] integrates purposeful robotic control with multi-modal perception via chain-of-thought reasoning. SwitchVLA [6] achieves task-switching success rates above 90%.

D. Next-Best-View Planning

Pred-NBV [1] learns to predict information gain of candidate viewpoints, observing up to 25.46% more surface points than conventional planners within the same temporal budget. GenNBV [7] generalises NBV planning across object categories via a policy network, optimising the flight path length versus coverage trade-off. Zhang et al. [10] demonstrate NBV-driven exploration with 3DGS-based reconstruction for semantic indoor mapping using autonomous UAVs.

E. Edge-Optimised Inference for Robotics

III. Semantic-Guided Autonomous Mapping

A. Closed-Loop Architecture

B. Edge Computing Deployment

Table I summarises the expected performance characteristics under cloud and edge deployment scenarios, based on LoRA fine-tuning and 8-bit quantisation strategies for VLA models.

TABLE I
Comparison of VLA Deployment Configurations

Configuration	Method	Latency (s)	VRAM (GB)
Cloud (Traditional)	Full fine-tuning	> 2.50	40–80
Edge (Optimised)	8-bit + LoRA	0.60–0.85	8–12

The 3DGS rendering pipeline achieves frame rates above 100 fps [11], while feed-forward reconstruction variants reduce per-scene latency to 0.29–0.63 s [11]. Zhang et al. [10] have demonstrated the feasibility of 3DGS-based semantic reconstruction on autonomous UAV platforms.

C. Semantic Uncertainty Modulation

IV. Comparative Analysis

This section synthesises quantitative performance metrics from the surveyed literature along four evaluation axes.

A. Reconstruction Quality Metrics

Key metrics from the literature:

- **PSNR:** 3DGS achieves up to 28.93 dB in urban scenes [12].
- **SSIM:** Structural similarity scores of 0.918 [3].
- **LPIPS:** Perceptual similarity as low as 0.056 [3].
- **ATE:** Absolute Trajectory Error for SLAM localisation accuracy.
- **CD:** Chamfer Distance for point cloud geometric fidelity.

B. Exploration Efficiency Metrics

Key metrics from the literature:

- **Coverage Rate / AUC:** Measures how rapidly the planner achieves full coverage.
- **Surface Point Gain:** Pred-NBV achieves 25.46% more observed points [1].
- **Flight Path Length vs. Coverage:** GenNBV optimises this trade-off across object categories [7].

C. Cognitive Performance Metrics

Key metrics from the literature:

- **Task Switching Success Rate:** SwitchVLA achieves $>$ 90% [6].
- **Spatial Reasoning Accuracy:** Qwen2.5-VL achieves 75.6%–86.3% on EmbSpatial [4]; Lumo-1 exhibits comparable performance [5].

D. Hardware Feasibility Metrics

Key metrics from the literature:

- **Rendering FPS:** 3DGS exceeds 100 fps (up to 200 fps) [11].
- **Reconstruction Latency:** Feed-forward methods achieve 0.29–0.63 s per scene [11].
- **VLA VRAM:** Cloud $\approx$ 40–80 GB $\rightarrow$ Edge $\approx$ 8–12 GB (8-bit + LoRA).
- **VLA Latency:** Cloud > 2.50 s $\rightarrow$ Edge 0.60–0.85 s.

V. Discussion

A. Latency–Accuracy Trade-offs on Edge Devices

B. Dynamic Environments and Scene Changes

C. Safety-Critical Certification

D. Transferability to Unstructured Environments

E. Benchmark Standardisation

F. Multi-Robot Cooperative Exploration

G. Long-Horizon Mission Planning

VI. Conclusion

This survey has reviewed the convergence of 3D Gaussian Splatting, Vision-Language-Action models, and predictive Next-Best-View planning toward a unified framework for semantic-guided autonomous mapping. The quantitative evidence consolidated from the literature demonstrates that the integration of VLA models optimised via LoRA with geometric systems such as 3DGS is not only computationally feasible on edge hardware (VRAM $\approx$ 8–12 GB, latency < 1 s) but imperative for the next generation of mobile robots and UAVs in autonomous inspection missions.

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